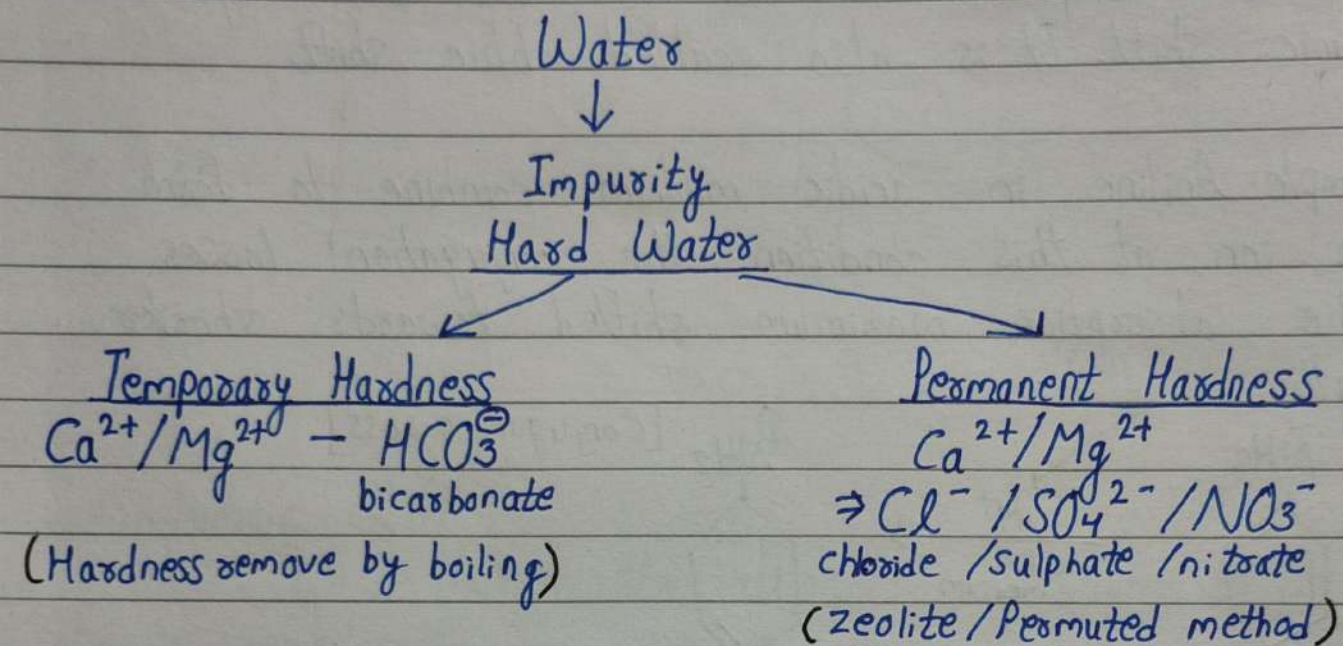


Module - IV: Water Technology

Sources & Impurities of Water



Hard Water

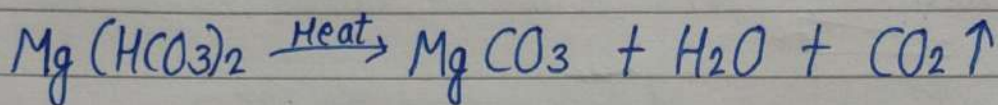
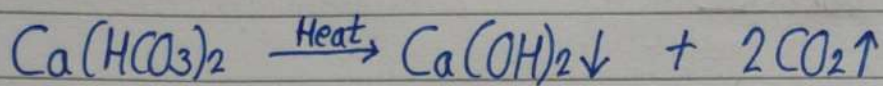
The water contain salts of calcium and magnesium of bicarbonates, chloride, sulphate, nitrate, etc, called hard water.

There are two types of hardness;

① Temporary Hardness:

The water contains ^{bicarbonate} ~~salts~~ of calcium and magnesium, called temporary hardness. It can be removed by boiling.

Bicarbonates of Calcium and Magnesium heated, carbonates and hydroxides of calcium and magnesium get precipitated.



② Permanent Hardness:

The water contains calcium and magnesium salts of chloride, sulphate, nitrate causes permanent hardness. It can not be removed by boiling and ^{can} only removed by zeolite or permuted method.

Softning of Water by Zeolite / Permuted Method

The removal of hardness causing agents like calcium and magnesium bicarbonates, nitrate, chloride, sulphate, etc, called as water softning.

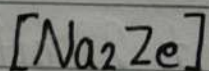
greek words

Zeolite is derived from λ zein and lithos, which means boiling stone. Zeolite is of two types;

- ① Natural zeolites are non-phosous green sand and are more durable.
- ② Synthetic or permuted zeolites are phosous and the general chemical formula is;

$$\text{Na}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot x\text{SiO}_2 \cdot y\text{H}_2\text{O} \quad (\text{Sodium Zeolite})$$

Sodium aluminium silicate



vImp.

Zeolite / Permutit Process

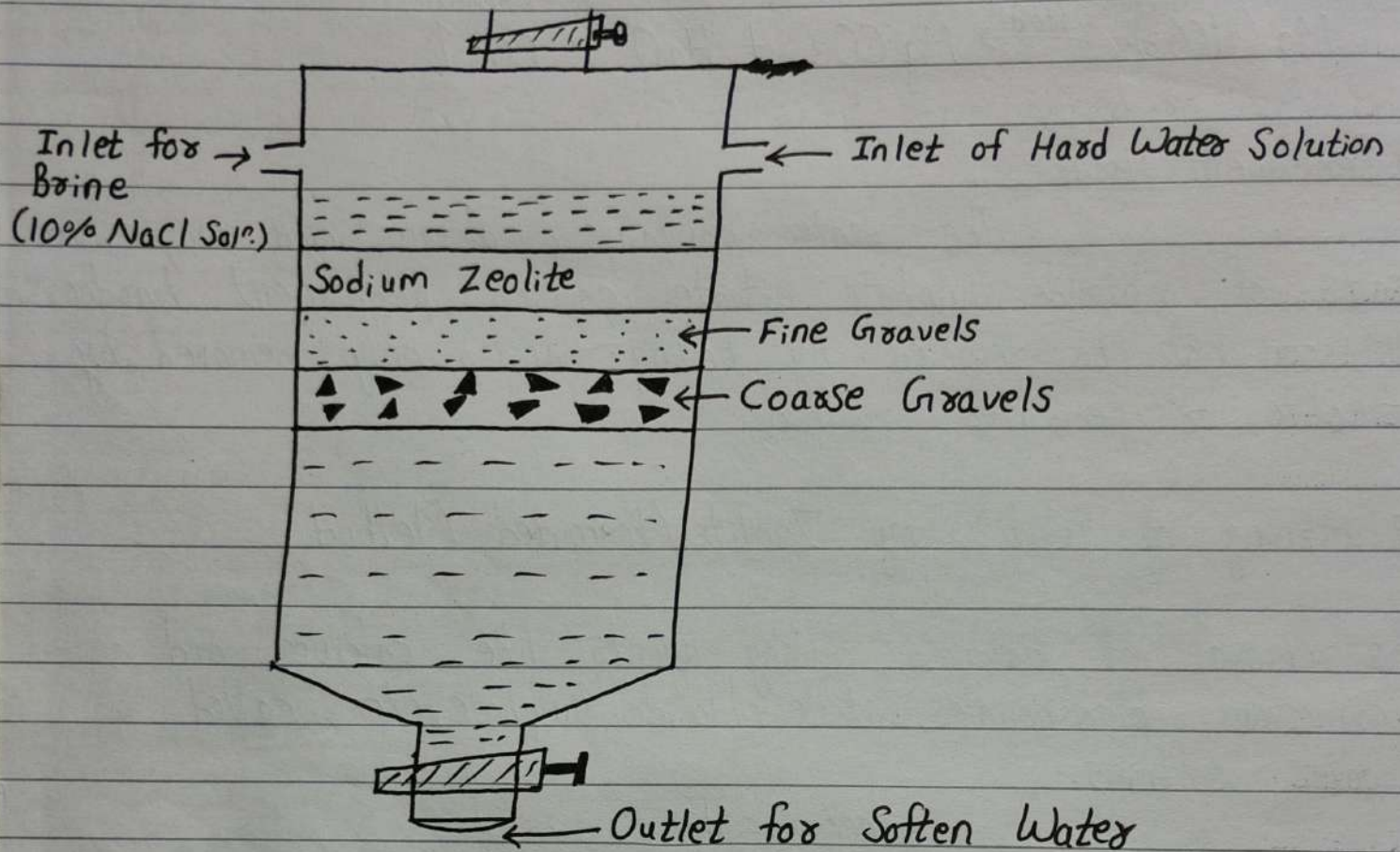
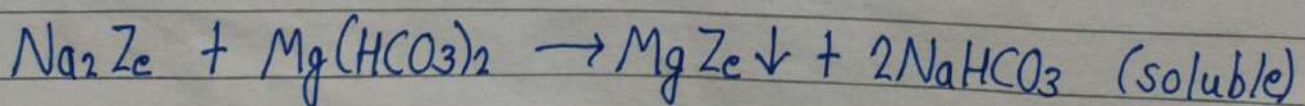
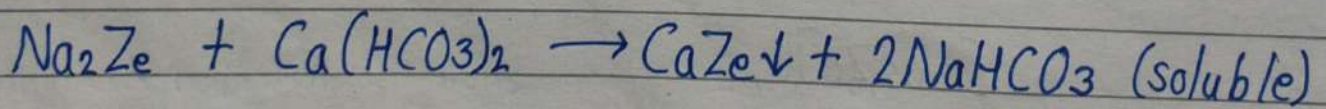
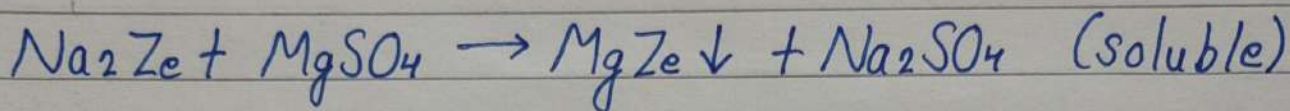
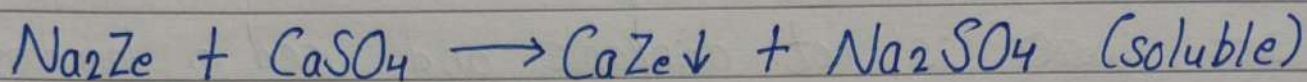
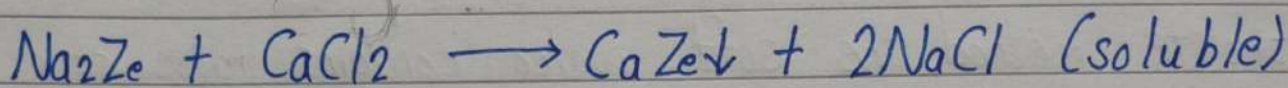
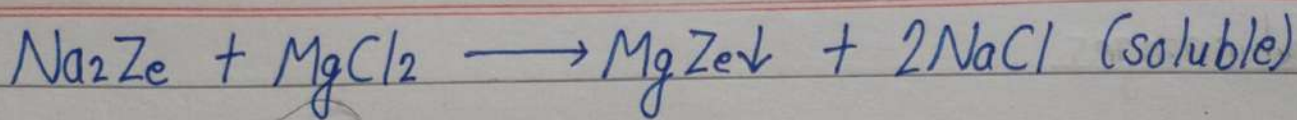


Fig: Zeolite Softner

Zeolites are complex compound of sodium, aluminium silicate, that are insoluble and ^{they} exchange sodium ion by calcium and magnesium ion, they are called as base exchanger.

Calcium and magnesium ions reacts with sodium zeolite. Reactions are as follows:

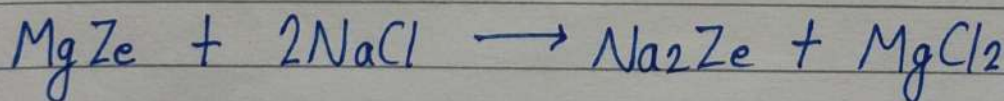
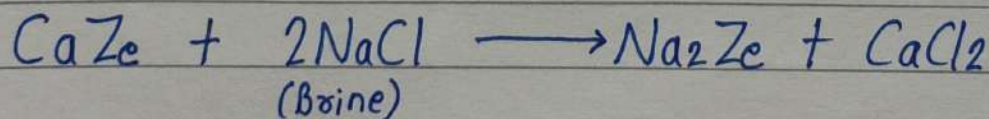




The zeolite softener consist of steel tank packed with a thick layer of zeolite and the hard water taken from the upper side of softener.

Now, the calcium and magnesium contact with sodium zeolite, the sodium zeolite get converted into calcium and magnesium zeolite and soluble sodium salts collected in bottom with soft water.

When the ^{sodium} zeolite is completely exhausted then it caused then it do not soft water. Now, at this condition 10% NaCl solution can added from upper side and regeneration process will start. Sodium zeolite regain again.



}

Regeneration

Advantages

- The equipment is compact and occupy small space.
- It requires less time for softening.
- It is cheap process due to regeneration.

Disadvantages

- By zeolite method, the soft water contains more conc. of sodium ion.
- The soft water is not drinkable and used in laundry.

$$1 \text{ mg/l} = 1 \text{ ppm} = 0.1^\circ \text{ French} = 0.07^\circ \text{ Clark}$$

Date _____

Page _____

* Numerical Based on Water Hardness

Constituents	Molar Mass	(Calcium carbonate Equivalent) Multiplication Factor
$\text{Ca}(\text{HCO}_3)_2$	162	$100/162$
$\text{Mg}(\text{HCO}_3)_2$	146	$100/146$
CaCl_2	111	$100/111$
MgCl_2	95	$100/95$
CaSO_4	136	$100/136$
MgSO_4	120	$100/120$

- Q) A water sample contains 408 mg of Calcium sulphate per litre. Calculate the hardness in terms of calcium carbonate equivalent. ($\text{Ca} = 40$, $\text{S} = 32$ & $\text{O} = 16$)

Solⁿ:

$$\text{Hardness } \text{CaSO}_4 = \frac{100}{136} \times 408 = 300 \text{ mg/l}$$

- Q) A water sample on analysis was found to contain following impurities in mg/l.
- $$\begin{aligned} \text{Ca}(\text{HCO}_3)_2 &= 10 \text{ mg/l} \\ \text{Mg}(\text{HCO}_3)_2 &= 8.5 \text{ mg/l} \\ \text{CaSO}_4 &= 22 \text{ mg/l} \end{aligned}$$

$$\text{MgSO}_4 = 24 \text{ mg/l}$$

Calculate the temporary, permanent and total hardness.

Solⁿ:

$$\text{Ca(HCO}_3)_2 = 10 \text{ mg/l} \Rightarrow \text{F.H}$$

Temporary Hardness

$$= \frac{100}{162} \times 10 + \frac{100}{146} \times 8.5$$

$$= 11.99 \text{ mg/l}$$

Hard Permanent Hardness

$$= \frac{100}{136} \times 22 + \frac{100}{120} \times 24$$

$$= 36.17 \text{ mg/l}$$

$$\therefore \text{Total Hardness} = 48.16 \text{ mg/l}$$

Q) Why does hard water consume a lot of soap?
Calculate the temporary, permanent and total hardness of water sample, which is analysed as:

$$\text{Ca}(\text{HCO}_3)_2 = 40.5 \text{ mg/l}$$

$$\text{Mg}(\text{HCO}_3)_2 = 36.5 \text{ mg/l}$$

$$\text{CaCl}_2 = 27.75 \text{ mg/l}$$

$$\text{NaCl} = 10 \text{ mg/l}$$

Solⁿ:

Temporary Hardness

$$= \frac{100}{162} \times 40.5 + \frac{100}{146} \times 36.5$$

$$= 50 \text{ mg/l}$$

Permanent Hardness

$$= \frac{100}{111} \times 27.75$$

$$= 25 \text{ mg/l}$$

$$\therefore \text{Total Hardness} = 50 + 25 = 75 \text{ mg/l}$$

⇒ In hard water, ester of soap do not decarboxylate, so it does not give foam. So, hard water require a lot of soap.

Q) Calculate the temporary, permanent and total hardness of the following water sample:

$$\text{Ca}(\text{HCO}_3)_2 = 4 \text{ ppm}$$

$$\text{Mg}(\text{HCO}_3)_2 = 6 \text{ ppm}$$

$$\text{MgSO}_4 = 8 \text{ ppm}$$

$$\text{CaSO}_4 = 10 \text{ ppm}$$

Convert it into $^\circ\text{French}$ & $^\circ\text{Clark}$.

Solⁿ:

Temporary Hardness

$$= \frac{100}{162} \times 4 + \frac{100}{146} \times 6$$

$$= 6.57 \text{ ppm}$$

Permanent Hardness

$$= \frac{100}{120} \times 8 + \frac{100}{136} \times 10$$

$$= 14.01 \text{ ppm}$$

$$\therefore \text{Total hardness} = 20.58 \text{ ppm}$$

$$^\circ\text{French}, 20.58 \times 0.1 = 2.058 \text{ }^\circ\text{French}$$

$$^\circ\text{Clark}, 20.58 \times 0.07 = 1.4406 \text{ }^\circ\text{Clark}$$